

Yuni Wu

South San Francisco, CA

yunihsuanwu@gmail.com | (720) 487-5403 | linkedin/wih | github/yuni-wyx | portfolio

Summary

Machine Learning Engineer focused on turning promising models into dependable systems. Passionate about improving model reliability through evaluation, monitoring, and validation so teams can confidently deploy and operate ML products.

Technical Skills

Languages: Python, SQL, Go, R

Machine Learning: CNNs/ResNet, Transformers, hierarchical classification, class imbalance handling, targeted oversampling

Evaluation: cross-validation, ROC-AUC, F1-score, calibration, ablation workflows, A/B testing, model benchmarking, MAE, R^2

ML Systems: Cloud Run inference evaluation, Pub/Sub, Cloud Storage, Docker, metadata pipelines, monitoring, regression validation

Data & APIs: FastAPI, REST APIs, BigQuery, PostgreSQL, ETL/ELT pipelines, schema validation, anomaly detection

Experience

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Engineered ROS/rosbag data pipelines to curate 97K+ state-action samples from Unitree Go1 trajectories.
- Benchmarked ResNet behavior-cloning policies across autonomous navigation and human-interaction scenarios.
- Evaluated VLM-guided navigation strategies for decision consistency, semantic understanding, and deployment robustness.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Developed a ResNet-based 22-class classification system on 50K+ samples, achieving 0.92 F1-score.
- Improved minority-class F1 by +30% through targeted oversampling, class-imbalance analysis, and validation-focused iteration.
- Designed cross-validation, ROC-AUC, calibration, and validation workflows to measure model reliability under class imbalance.
- Evaluated deployment tradeoffs through ablation analysis, reducing inference latency 5x from 0.15s to 0.03s per image.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered SQL validation and anomaly-detection pipelines for 800+ production scenarios, reducing manual QA by 80%.
- Built regression monitoring pipelines to surface data inconsistencies faster, accelerating reporting speed by 78%.
- Optimized distributed ETL workflows and GCP data-access patterns, reducing query latency by 15%.

Projects

NailAI — Cloud-Native Inference and Monitoring Platform

[demo]

- Built Cloud Run, Pub/Sub, and Cloud Storage workflows for asynchronous image processing and retry-safe orchestration.
- Designed request, job, and prediction metadata pipelines for querying, monitoring, downstream analytics, and reliability review.
- Achieved 0% failure at 200 concurrent users with p95 latency of 6.8s through idempotent orchestration and retries.
- Optimized BigQuery partitioning and clustering for downstream analysis, warehouse tracking, and reduced query latency.

VLM-Based Social Navigation for Quadruped

[project]

- Built ROS/rosbag telemetry pipelines to transform raw robot data into structured state-action datasets for evaluation.
- Constructed an offline evaluation dataset with 2.4K+ labeled state-action sequences across 13 robot-navigation scenarios.
- Trained ResNet behavior-cloning policies on 97K+ samples, achieving 0.11 MAE and $R^2 = 0.62$.

EmotionFlow — Applied ML Evaluation and Recommendation Quality

[demo]

- Designed LLM evaluation pipelines and prompt optimization workflows for recommendation and response quality analysis.
- Reduced hallucination by 34% through feedback-driven evaluation loops using preference signals and response analysis.
- Increased recommendation acceptance to 57% through A/B testing and iterative prompt optimization.

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0

May 2026

Soochow University — BS, Data Science | GPA: 3.7/4.0

June 2021